# Kinetic Modelling of Esterification of Ethanol with Sulfuric Acid

**Project Completion Report**

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# 1 Introduction

Diethyl sulfate (DES) is a significant industrial chemical used primarily as an alkylating agent in various organic synthesis processes. Its applications range from pharmaceuticals and agrochemicals to dyes and plasticizers, making it an essential substance in the chemical industry. However, traditional methods of DES production are often challenged by high production costs, complex process configurations, and the need for stringent operating conditions, such as elevated temperatures and pressures. These challenges motivate the development of more efficient and cost-effective production methods.

The conventional production of DES typically involves the reaction of ethanol with sulfur trioxide or chlorosulfonic acid, requiring specialized equipment and sophisticated control to ensure safe and efficient production. These processes not only involve multiple intricate steps but also generate significant quantities of by-products that necessitate careful handling and disposal. Such factors contribute to both the economic and environmental costs associated with DES production, driving a need for innovation in reaction pathways.

A promising alternative approach for DES production is the modified ethanol-sulfuric acid method, which simplifies the production process while lowering costs. In this modified method, DES is synthesized in two primary steps: first, the formation of ethyl hydrogen sulfate (EHS) from ethanol and sulfuric acid, with the assistance of anhydrous sodium sulfate as a drying agent, and second, the distillation of EHS under vacuum conditions to yield DES. Notably, the distillation step is a well-established procedure, and thus, this study focuses on the first step, which presents significant opportunities for optimization.

The formation of EHS is a crucial intermediate step in the production of DES, and understanding the kinetics of this reaction can offer valuable insights into improving yield and reducing production costs. In the first step, ethanol reacts with sulfuric acid to form EHS, a reaction that is highly influenced by factors such as temperature, the presence of water, and the concentration of reactants. Since water is generated as a by-product during the reaction, its removal becomes essential to shift the equilibrium towards the desired product. The use of anhydrous sodium sulfate as a drying agent helps in this regard, but further optimization can be achieved by exploring other adsorbents or varying reaction conditions.

This study aims to elucidate the reaction kinetics of EHS formation, providing a detailed analysis of the rate constants. The broader goal is to develop a comprehensive understanding of the process dynamics that can inform improvements in DES production, ultimately making the process more efficient, economical, and environmentally friendly.

# 2 Hypothesis

Diethyl sulfate (DES) production traditionally involves high-cost and complex methods requiring strict operating conditions. A modified ethanol-sulfuric acid process allows DES synthesis in two steps: (1) formation of ethyl hydrogen sulfate (EHS) from ethanol and sulfuric acid, with anhydrous sodium sulfate as the drying agent and (2) distillation of EHS under vacuum to yield DES. Since the distillation of EHS to produce DES is a well established step, this study focuses on the first step, aiming to investigate the reaction kinetics of EHS formation reaction under controlled conditions. The study aims to determine the rate constants, analyze the effect of change in initial mole ratio.

# 3 Proposed Methodology

## 3.1 Reaction Mechanism

The reaction between ethanol and sulfuric acid to form EHS can be represented by the following equation:

$$C_2H_5OH + H_2SO_4 \rightarrow C_2H_5HSO_4 + H_2O$$

In this reaction, EHS is produced along with water as a byproduct. To ensure maximal yield, water formed during the reaction is removed by reactive adsorption, using anhydrous sodium sulfate when necessary to further drive the reaction forward.

## 3.2 Materials Required

- **Reagents:** Ethanol ($C_2H_5OH$), Sulfuric Acid ($H_2SO_4$), Barium Chloride ($BaCl_2$), Anhydrous Sodium Sulfate ($Na_2SO_4$)
- **Apparatus:** Heating mantle with temperature control, Magnetic stirrer and stir bar, Pipettes, Three-neck round-bottom flask (1000 mL)

## 3.3 Procedure

1. **Reactor setup:** Used a 1000 mL glass stirred batch reactor with three necks. Equipped with a reflux condenser, thermometer and sample collection neck. Reactor placed in a constant-temperature bath with cooling water.
2. **Preparation of reactants:** Ethanol (100 mL) and anhydrous sodium sulfate added to the reactor. Sulfuric acid (100 mL) cooled separately to 5°C - 7°C.
3. **Reaction initiation:** Ethanol was placed in the batch reactor, and sulfuric acid was gradually added with continuous stirring. The reactor was maintained at room temperature under isothermal conditions.
4. **Sampling:** 1 mL samples taken at intervals of 4 minutes each.

# 4 Experimental Mishap and Remediation

During the drying stage of the $BaSO_4$ precipitate, we initially followed staff advice to use a furnace. However, instead of drying, the sample ignited. Upon further review, it was identified that both $H_2SO_4$ and ethanol—components in the experiment—are flammable, and thus, furnace drying poses a significant risk of combustion.

To remedy this, we switched to a safer drying method. We used multiple layers of filter paper and allowed the precipitate to dry under ambient room temperature for approximately three hours. This method effectively dried the precipitate without further incident, allowing us to proceed with weighing.

# 5 Post-Experimental Analysis

The post-experimental analysis involves a detailed examination of the experimental data obtained from the formation of ethyl hydrogen sulfate (EHS). This section starts by plotting the raw data, followed by fitting the data to different kinetic models: first-order reversible, first-order irreversible, and second-order reversible rate laws.

| Time (min) | Mass $BaSO_4$ | Ca | Cb | Cc |
|---|---|---|---|---|
| 0 | 8.519 | 9.941 | 9.371 | 0.0000 |
| 5 | 7.260 | 8.471 | 6.412 | 2.069 |
| 10 | 6.737 | 7.862 | 5.889 | 2.592 |
| 15 | 5.657 | 6.601 | 4.809 | 3.672 |
| 20 | 5.480 | 6.394 | 4.632 | 3.850 |
| 25 | 5.889 | 5.046 | 4.198 | 4.283 |
| 30 | 5.514 | 4.726 | 3.878 | 4.603 |
| 35 | 5.192 | 4.449 | 3.601 | 4.880 |
| 40 | 5.476 | 4.693 | 3.845 | 4.636 |
| 45 | 5.247 | 4.497 | 3.649 | 4.832 |
| 50 | 4.833 | 4.142 | 3.293 | 5.187 |
| 55 | 4.758 | 4.078 | 3.230 | 5.251 |
| 60 | 4.537 | 3.888 | 3.040 | 5.441 |

Table 1: Concentration and Mass of Data Over Time for M=1.1

| Time (min) | Mass $BaSO_4$ | Ca | Cc | Cb |
|---|---|---|---|---|
| 0 | 2.888 | 2.470 | 0.000 | 14.873 |
| 4 | 2.540 | 2.173 | 0.297 | 14.576 |
| 8 | 2.123 | 1.816 | 0.654 | 14.219 |
| 12 | 1.767 | 1.512 | 0.959 | 13.915 |
| 16 | 1.478 | 1.264 | 1.206 | 13.667 |
| 20 | 1.326 | 1.134 | 1.336 | 13.537 |
| 24 | 1.261 | 1.079 | 1.392 | 13.482 |
| 28 | 1.043 | 0.892 | 1.578 | 13.295 |
| 32 | 0.801 | 0.685 | 1.785 | 13.088 |
| 36 | 0.869 | 0.743 | 1.727 | 13.146 |
| 40 | 0.800 | 0.684 | 1.786 | 13.087 |
| 44 | 0.618 | 0.529 | 1.941 | 12.932 |
| 48 | 0.573 | 0.490 | 1.980 | 12.893 |
| 52 | 0.388 | 0.332 | 2.138 | 12.735 |
| 56 | 0.289 | 0.248 | 2.223 | 12.651 |
| 60 | 0.273 | 0.233 | 2.237 | 12.636 |

Table 2: Concentration and Mass of Data Over Time for M=6

## 5.1 Plotting Raw Data

In order to plot the raw data, we use MATLAB to define the values for the axes.

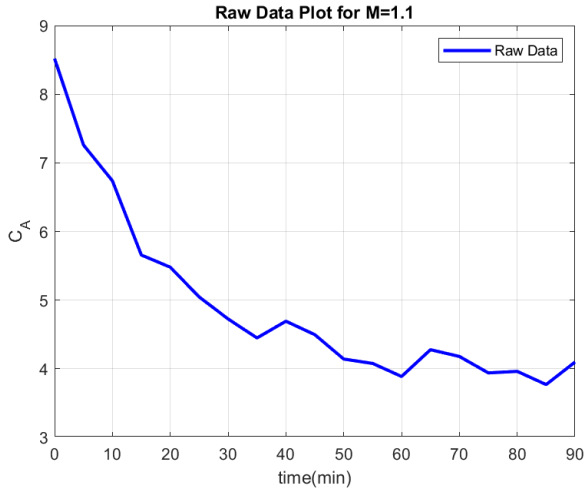


Figure 1: Raw data for MR=1.1

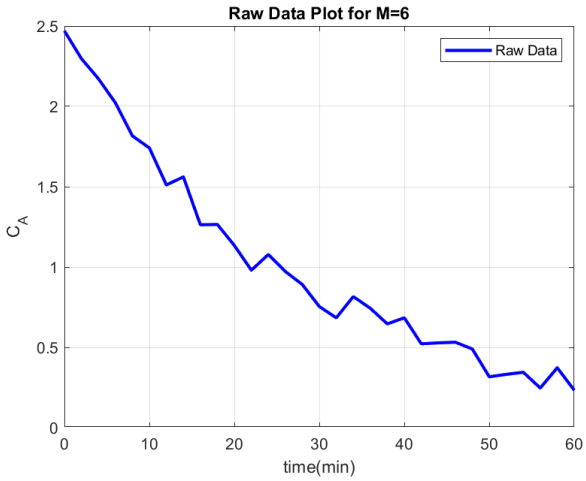


Figure 2: Raw Data for MR=6

```
% Plotting the Raw Data
% Data for MR = 1.1
data1 = [0, 8.519; 5, 7.2597; 10, 6.7374; 15, 5.6569; 20, 5.4798; 25, 5.0464; ...
         30, 4.7257; 35, 4.4491; 40, 4.6929; 45, 4.4966; 50, 4.1415; 55, 4.0776; ...
         60, 3.8877; 65, 4.2778; 70, 4.1793; 75, 3.9400; 80, 3.9621; 85, 3.7699; 90, 4.0980];
t1 = data1(:,1);C_A1 = data1(:,2);
% Plot for MR = 1.1
figure(1);
plot(t1, C_A1, 'b-', 'LineWidth', 2, 'DisplayName', 'MR = 1.1');
xlabel('Time (min)');
ylabel('C_{A}');
title('Raw Data Plot for MR = 1.1');
grid on;
legend;
% Data for MR = 6
data2 = [0, 2.470; 2, 2.297; 4, 2.172; 6, 2.022; 8, 1.816; 10, 1.741; ...
         12, 1.511; 14, 1.561; 16, 1.263; 18, 1.265; 20, 1.134; 22, 0.980; ...
         24, 1.078; 26, 0.972; 28, 0.891; 30, 0.754; 32, 0.684; 34, 0.816; ...
         36, 0.743; 38, 0.646; 40, 0.684; 42, 0.522; 44, 0.528; 46, 0.532; ...
         48, 0.490; 50, 0.317; 52, 0.332; 54, 0.345; 56, 0.247; 58, 0.373; 60, 0.233];
t2 = data2(:,1);C_A2 = data2(:,2);
% Plot for MR = 6
figure(2);
plot(t2, C_A2, 'b-', 'LineWidth', 2, 'DisplayName', 'MR = 6');
xlabel('Time (min)');
ylabel('C_{A}');
title('Raw Data Plot for MR = 6');
grid on;
legend;
```

Listing 1: MATLAB Code for above graphs

### First Order Irreversible Reaction Fitting

Next, we fit the data to a first-order irreversible reaction model to determine if this simpler kinetic model can adequately describe the reaction. We find that for MR=6 the reaction rate law follows an apt first order kinetics.

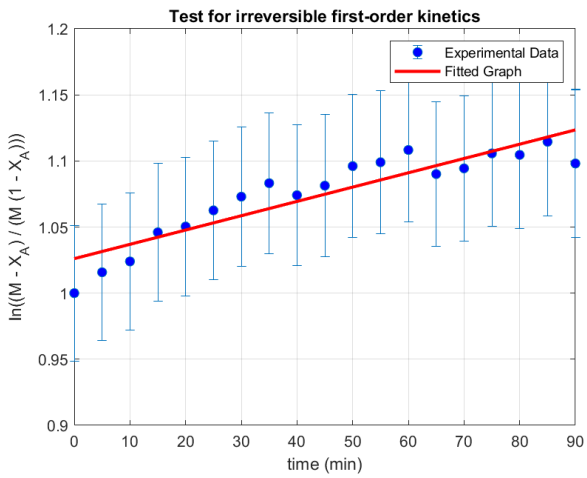


Figure 3: Fitted data for MR=1.1 w.r.t $C_a$

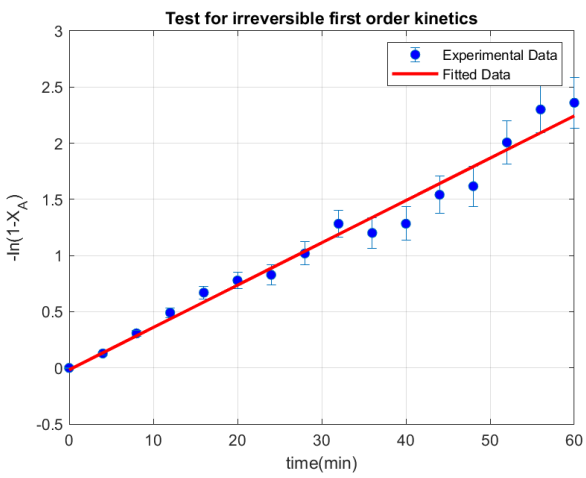


Figure 4: Fitted Data for MR=6 w.r.t $X_a$

```
%m=6 first order
clear all;
clc;
%Data
lnX = [0, 0.1283, 0.3077, 0.4912, 0.6701, 0.7785, 0.8285, 1.0187, 1.2829,
    1.2013, 1.2838, 1.5412, 1.6173, 2.0068, 2.3003, 2.3597];
Ca = [2.4702, 2.1728, 1.8160, 1.5115, 1.2640, 1.1341, 1.0787, 0.8919, 0.6848,
    0.7430, 0.6842, 0.5290, 0.4902, 0.3320, 0.2476, 0.2333];
t = [0, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60];

% Fit a linear curve to lnX vs t
coeff_first_order = polyfit(t, lnX, 1);

% Generate fitted values for lnX
ln_X_fitted = polyval(coeff_first_order, t);

% Generate fitted values for Ca using the exponential model
C_A_fitted = Ca(1) * exp(-coeff_first_order(1) * t);

% Calculate a constant error for demonstration (e.g., 5% of each fitted value)
lnX_error = 0.05 * ln_X_fitted; % 5% error for lnX fitted values
Ca_error = 0.05 * C_A_fitted;   % 5% error for Ca fitted values

%% Plot lnX with error bars
figure(1);
errorbar(t, lnX, lnX_error, 'o', 'MarkerFaceColor', 'b', 'DisplayName', '
    Experimental Data');
hold on;
plot(t, ln_X_fitted, 'r-', 'LineWidth', 2, 'DisplayName', 'Fitted Data');
xlabel('time(min)');
ylabel('-ln(1-X_{A})');
title('Test for irreversible first order kinetics');
grid on;
legend;
```

```
hold off;

% Display the rate constant
fprintf('First-order rate constant k = %.4f min^{-1}\n', coeff_first_order(1));
```

Listing 2: MATLAB Code

### Second Order irreversible Reaction Fitting

Moving on, after getting an apt fit for MR = 6, we apply the second-order irreversible reaction model to determine if the kinetics of EHS formation are better described by a higher-order mechanism,for MR = 1.1. This analysis provides a more comprehensive understanding of the reaction dynamics.

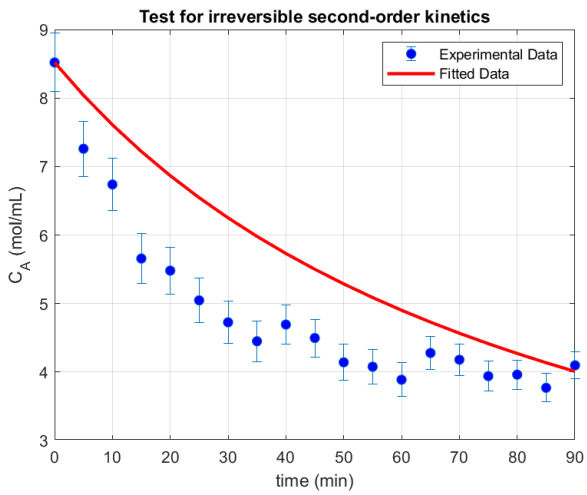


Figure 5: Fitted data for MR=1.1 w.r.t $C_a$

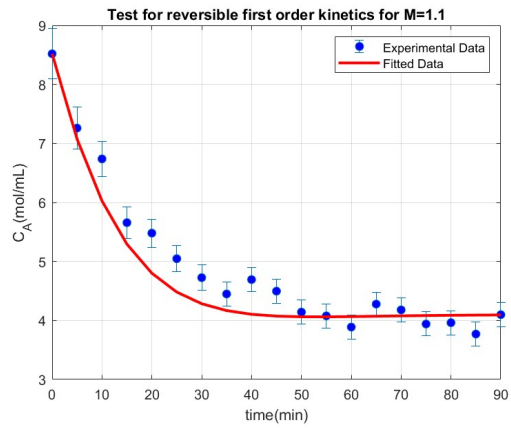


Figure 6: First Order Reversible for M=1.1

```
% Data
ln_XA = [1; 1.0158; 1.0240; 1.0460; 1.0504; 1.0626; 1.0730; 1.0832; 1.0741; 1.0813; ...
      1.0961; 1.0990; 1.1083; 1.0901; 1.0944; 1.1057; 1.1046; 1.1145; 1.0981];
t = [0; 5; 10; 15; 20; 25; 30; 35; 40; 45; 50; 55; 60; 65; 70; 75; 80; 85; 90];

% Fit a linear curve to ln(C_A) vs t
coeff_first_order = polyfit(t, ln_XA, 1);
fitted = polyval(coeff_first_order, t);
ln_CA_error = 0.05 * fitted;
k = coeff_first_order(1) / (0.1 * 8.519);

% Concentration Data
Ca = [8.519; 7.2597; 6.7374; 5.6569; 5.4798; 5.0464; 4.7257; 4.4491; 4.6929; ...
      4.4966; 4.1415; 4.0776; 3.8877; 4.2778; 4.1793; 3.9400; 3.9621; 3.7699; 4.0980];
fitted2 = coeff_first_order(1) .* t;
Xa = 1.1 .* (exp(fitted2) - 1) ./ ((1.1) .* exp(fitted2) - 1);
Ca_fit = 8.519 .* (1 - Xa);
Ca_error = 0.05 * Ca_fit;

% Plot 1
figure(1);
```

```
errorbar(t, ln_XA, ln_XA_error, 'o', 'MarkerFaceColor', 'b', 'DisplayName', '
    Experimental Data');
hold on;
plot(t, fitted, 'r-', 'LineWidth', 2);
xlabel('time (min)');
ylabel('ln((M - X_{A}) / (M (1 - X_{A})))');
title('Test for irreversible second-order kinetics');
grid on;
legend('Experimental Data', 'Fitted Graph');
hold off;

% Plot 2
figure(2);
errorbar(t, Ca, Ca_error, 'o', 'MarkerFaceColor', 'b', 'DisplayName', '
    Experimental Data');
hold on;
plot(t, Ca_fit, 'r-', 'LineWidth', 2, 'DisplayName', 'Fitted Data');
xlabel('time (min)');
ylabel('C_{A} (mol/mL)');
title('Test for irreversible second-order kinetics');
grid on;
legend('Experimental Data', 'Fitted Data');
hold off;
```

Listing 3: MATLAB Code for Loss Analysis

### Second Order reversible Reaction Fitting

Since, the results were not accurate enough for the second order irreversible kinetics, the data might show a better result when compared to reversible kinetics, since the concentration converges to an offset with respect to 0.

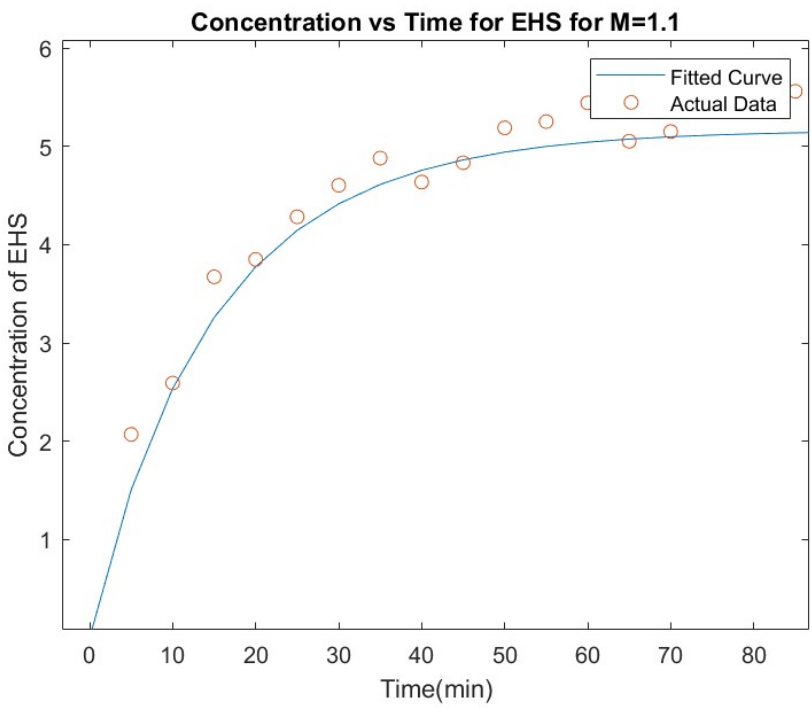


Figure 7: First Order reversible second order fit for MR=1.1

# 6 Summary

The kinetic analysis below offers a thorough understanding of the esterification reaction between ethanol and sulfuric acid to form ethyl hydrogen sulfate (EHS). The major findings are summarized below:

### Reaction Behavior at Different Molar Ratios

- **1.1:1 Molar Ratio (Ethanol to Sulfuric Acid)**
  - The experimental data for this ratio did not align with irreversible first-order or second-order kinetics.
  - A reversible second-order kinetic model, however, provided a good fit, as seen in the linear plots of the rate equation versus time, suggesting that a reversible second-order mechanism governs the reaction at this molar ratio.
  - This implies that equilibrium significantly influences the yield of the ester product under these conditions.
- **6:1 Molar Ratio (Ethanol to Sulfuric Acid)**
  - At this higher molar ratio, the reversible second-order rate law function versus time did not provide a linear fit.
  - An irreversible first-order kinetic model, however, demonstrated a strong linear correlation, validating an irreversible first-order rate law for the reaction at this molar ratio.
  - This was further supported by differential analysis, with $\ln(-r_A)$ versus $\ln(C_A)$ plots reinforcing the applicability of irreversible first-order kinetics.

### Mechanistic Variation

The reaction mechanism shifts based on reactant molar ratios:

- **1.1:1 Ratio** - Follows a reversible second-order kinetics, indicating an elementary mechanism driven by equilibrium.
- **6:1 Ratio** - Adopts a non-elementary mechanism best described by irreversible first-order kinetics, suggesting a unimolecular rate-determining step.

This shift underscores a multiplicity of mechanisms, with higher ethanol concentrations favoring a simpler, irreversible mechanism..

# 7 Proposed Mechanism

**Assumptions**

The assumptions are based partly on experimental information and partly on accumulated knowledge about characteristics of the different species involved in the reaction. The various assumptions involved are:

1. The alcohol forms a complex $Q$ with a component of the acid.
2. A small proportion of the complex gets protonated.
3. The rate-determining step is a reversible, unimolecular decomposition of the protonated complex to form the ester, since experimental observation indicates an irreversible first-order reaction.

From the above-stated assumptions, the following steps are postulated:

$$\mathrm{H_2SO_4} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \mathrm{HSO_4^- + H^+} \quad \text{(fast)}$$

$$\mathrm{C_2H_5OH + HSO_4^-} \underset{k_5}{\overset{k_3}{\rightleftharpoons}} \mathrm{Q} \quad \text{(fast)}$$

$$\mathrm{Q + H^+} \underset{k_5}{\overset{k_4}{\rightleftharpoons}} \mathrm{QH^+} \quad \text{(fast)}$$

$$\mathrm{QH^+} \xrightarrow{k_6} \mathrm{C_2H_5HSO_4 + H_2O} \quad \text{(slow)}$$

Considering the slowest step among the above as the rate-determining one, the rate equation takes the form

$$r_{\mathrm{EHS}} = k_6[QH^+]$$

Employing the pseudo-steady state approximation, the concentration of the intermediate $[QH^+]$ is expressed in terms of $C_2H_5OH$ and $HSO_4^-$. Applying the pseudo-steady state approximation for all intermediates:

$$[QH^+] = \frac{k_3}{k_6}[C_2H_5OH][HSO_4^-] \tag{6}$$

$$[HSO_4^-] = \frac{k_1[H_2SO_4]}{k_2[H^+] + k_3[C_2H_5OH]} \tag{7}$$

Therefore,

$$r_{\mathrm{EHS}} = k_a \frac{[C_2H_5OH][H_2SO_4]}{[H^+] + k_b[C_2H_5OH]} \tag{8}$$

where $k_a = k_1 \cdot \frac{k_3}{k_2}$ and $k_b = \frac{k_3}{k_2}$.

Since at high mole ratios, $[C_2H_5OH] \gg [H^+]$, the above equation gets simplified to the following form:

$$r_{\mathrm{EHS}} = k[H_2SO_4] \tag{9}$$

where $k = \frac{k_a}{k_b}$.

This rate equation is the same as the one obtained from the analysis of the rate data by both integral and differential methods. Hence, the assumed mechanism is consistent with the experimental observations.

# 8 Possible sources of Errors

- **Measurement Inaccuracy:** Slight inaccuracies in measuring volumes or masses of reactants can lead to deviations in the expected results.
- **Moisture Presence:** Moisture in the environment or reactants can affect the weight and concentration measurements, leading to inaccuracies.
- **Temperature and Stirring Uniformity:** Inconsistent temperature or stirring can result in uneven reaction conditions, impacting the reaction rate and yield.
- **Sampling Errors:** Variability in the timing or handling of samples during the reaction may introduce inconsistencies in the data collected.
- **Evaporation of Volatile Components:** If volatile reactants like ethanol are used, slight evaporation may occur, affecting concentration measurements.

## Contributions of Team Members

1. **Alankrit Gupta (220110):**
   - Led the experimental setup for Molar Ratio (MR) 1.1, ensuring proper integration of all related datasets.
   - Analyzed initial results and compiled inferences, confirming the continuation of the proposed reaction mechanism.
   - Performed detailed calculations for fitting parameters across different reaction orders using differential methods.
2. **Aniket Shivhare (221022):**
   - Developed and proposed the reaction mechanism for MR=6, providing a comprehensive understanding of the system.
   - Led the fitting of parameters for multiple reaction orders specific to MR=6, ensuring accuracy in modeling.
   - Conducted extensive data modeling calculations, ensuring robust analysis of the dataset.
3. **Aryan Nigam (220227):**
   - Engaged in extensive data modeling for MR=1.1, fitting experimental data into appropriate models and deriving fitting curves.
   - Identified and determined key parameters for the fitting process, ensuring accuracy in model predictions.
   - Contributed significantly to the experimental setup by selecting and assembling the necessary apparatus.
4. **Ayush Bokad (220299):**
   - Proposed and evaluated various reaction orders for different molar ratios, contributing to the experiment's design.
   - Developed methods for continuous kinetic modeling, providing critical insights into the system's behavior.
   - Assisted in verifying experimental results, ensuring their consistency with theoretical predictions.
5. **Diya Saraf (220383):**
   - Contributed to the graphical modeling of experimental results, aiding in the visualization of data trends.
   - Assisted in the development and refinement of data models, improving the overall quality of analysis.
   - Helped organize and manage experimental data, ensuring a streamlined process for data analysis.
6. **Mitesh Wandhare (220650):**
   - Proposed the experimental mechanism prior to conducting the experiment, laying a solid foundation for the study.
   - Played a key role in building and configuring the experimental setup, particularly in ensuring proper temperature control.

- Provided essential contributions to data modeling, aiding in the accurate interpretation of experimental results.

7. **Sanskar Srivastava (220967):**
   - Assisted in the proposal of the initial reaction mechanism for MR=6 and identified two defining molar ratios.
   - Contributed to experimental modeling efforts, particularly focusing on various reaction orders in MR=1.1.
   - Verified experimental equations post-experiment and identified suitable integrators to ensure the validity of results.
8. **Shrimi Agrawal (221036):**
   - Determined experimental parameters, such as volume and other critical factors, to ensure accurate experimental conditions.
   - Assisted in the graphical representation of experimental results, contributing to data visualization.
   - Played a crucial role in fitting the correct parameters into the experimental equations, ensuring model accuracy.

## Additional Contributions

In addition to the individual contributions listed above, all team members made significant contributions to the following aspects of the project:

- Participated in the experimental search, contributing to the identification of appropriate reaction conditions.
- Collaborated on the setup management, ensuring the proper configuration and functioning of the experimental environment.
- Contributed to report preparation, proofreading, and maintaining safety protocols throughout the experiment.

**Overall, each member contributed equally to the success of the experiment.**

Alonkrit Gupta

Aniket Shivhare

Aryan Nigam

Mitesh Wandhare

Ayush Bakad

Sanskaar Srivastava

DIYA SARAF

Shrima Agrawal